Lean 4 Jacobian-Ring Certificates for the Fermat Quintic Fourfold: Smoothness, Hilbert Series $(1+t+t^2+t^3)^6$, and Primitive Hodge Numbers $(0, 120, 580, 120)$

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Abstract

For $G = x_0^5 + \cdots + x_5^5$ we give machine-checked (Lean 4, Mathlib) proofs that the partial derivatives $\partial_i G = 5x_i^4$ have no common zero away from the origin in characteristic 0, so $X = V(G) \subset \mathbb{P}^5$ is smooth, and we compute in Lean the Hilbert series $(1+t+t^2+t^3)^6$ of the Jacobian ring $R = \mathbb{C}[x]/(x_0^4, \ldots, x_5^4)$: 19 coefficients summing to 4096, with $R_4 = 120$, $R_9 = 580$, $R_{14} = 120$. Via Griffiths' residue theorem (cited) these are the primitive Hodge numbers $h^{3,1}, h^{2,2}_{\text{prim}}, h^{1,3}$, and $h^{4,0} = 0$. We explain how the series is encoded as a decidable list computation, cross-check it against the Euler characteristic and an independent Singular computation, and apply the same Jacobian-ring analysis to the Dwork-type members $\psi = 1$ (singular, Lean) and $\psi = 2$ (smooth, Lean). All declarations use only the standard axioms; no claim about algebraic cycles beyond cited results is made.

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1 Introduction

The Jacobian ring is where the Hodge theory of a smooth hypersurface becomes finite algebra. For the Fermat quintic fourfold this algebra is explicit, which makes it a clean target for formal verification. Earlier versions of this record contained a two-page note certifying the partial-derivative formulas (1.0.1, 1.1.0) and a short update note (1.2.0). This version is the full paper and supersedes them. The focus here is the Jacobian ring itself: its structure, the formal computation of its Hilbert series, and exactly where Lean stops and Griffiths' theorem begins.

2 Setup and smoothness

Let K be a field of characteristic 0 (for Hodge theory, $K = \mathbb{C}$) and

$$G = \sum_{i=0}^{5} x_i^5 \in K[x_0, \dots, x_5], \qquad X = V(G) \subset \mathbb{P}^5_K.$$

G is homogeneous of degree 5 (Lean: `QuinticHost.G_isHomogeneous`), so X is a hypersurface of dimension 4. Euler's identity $5G = \sum_i x_i\, \partial_i G$ (Lean: `FermatQuintic.euler_homogeneous`) shows that a common zero of the partials lies on X.

Theorem 2.1 (Lean: `QuinticHost.pderiv_G`, `QuinticHost.smooth`)**.** *$\partial_i G = 5x_i^4$, and if $\operatorname{char} K = 0$ the six partials vanish simultaneously only at $x = 0$. Consequently X is smooth (Jacobian criterion).*

Proof. $5x_i^4 = 0$ with $5 \neq 0$ in K gives $x_i^4 = 0$, hence $x_i = 0$ in the field K. □

The Jacobian criterion itself (a projective hypersurface is smooth iff the partials have no common zero off the origin) is standard and is not formalised; Lean proves the polynomial statement.

Basic invariants: $K_X = \mathcal{O}_X(5-6) = \mathcal{O}_X(-1)$, so X is Fano of index 1. By Lefschetz, $b_0 = b_2 = b_6 = b_8 = 1$ and $b_1 = b_3 = b_5 = b_7 = 0$. The topological Euler characteristic of a smooth degree-d hypersurface in $\mathbb{P}^5$ is $\chi = \frac{(1-d)^6-1}{d} + 6$; for $d = 5$ this is $\frac{4096-1}{5} + 6 = 825$, so $b_4 = 825 - 4 = 821$ and the primitive part has dimension 820.

3 The Jacobian ring and its Hilbert series

The Jacobian ideal is $J = (\partial_0 G, \dots, \partial_5 G) = (x_0^4, \dots, x_5^4)$ and the Jacobian ring is

$$R = \mathbb{C}[x_0, \dots, x_5]/J \;=\; \bigotimes_{i=0}^{5} \mathbb{C}[x_i]/(x_i^4).$$

Since $x_0^4, \dots, x_5^4$ is a regular sequence, R is a graded Artinian complete intersection with monomial basis $\{x^a : 0 \le a_i \le 3\}$, so

$$H_R(t) = \Big(\frac{1-t^4}{1-t}\Big)^6 = (1+t+t^2+t^3)^6, \qquad \dim R = 4^6 = 4096,$$

with socle in degree $6 \cdot 3 = 18$ and the symmetry $\dim R_k = \dim R_{18-k}$ (Macaulay duality). For an arbitrary smooth quintic in six variables the partials still form a regular sequence of six forms of degree 4, so the same Hilbert series holds (cited, not formalised). For example R_4 is spanned by

k	0	1	2	3	4	5	6	7	8	9
$\dim R_k$	1	6	21	56	**120**	216	336	456	546	**580**
k	10	11	12	13	14	15	16	17	18	
$\dim R_k$	546	456	336	216	**120**	56	21	6	1	

Table 1: Coefficients of $(1+t+t^2+t^3)^6$. Sum = 4096. Bold: the Griffiths degrees.

degree-4 monomials in six variables with all exponents ≤ 3: $\binom{9}{5} - 6 = 126 - 6 = 120$ (only the six x_i^4 are excluded).

4 Griffiths residues and the Hodge numbers

Theorem 4.1 (Griffiths [1], cited)**.** *For a smooth hypersurface $X = V(G) \subset \mathbb{P}^{n+1}$ of degree d, $n = \dim X$,*

$$H^{n-q,q}_{\mathrm{prim}}(X) \;\cong\; R_{d(q+1)-(n+2)}.$$

Here $n = 4$, $d = 5$, so $h^{4-q,q}_{\mathrm{prim}} = \dim R_{5q-1}$:

q	degree $5(q+1)-6$	$h^{4-q,q}_{\mathrm{prim}}$	Lean
0	-1	0	`quintic_h40_zero`
1	4	120	`quintic_primitive_hodge`
2	9	580	`quintic_primitive_hodge`
3	14	120	`quintic_primitive_hodge`
4	19	0	(beyond socle degree 18)

The total $0+120+580+120+0 = 820$ agrees with the Euler-characteristic computation of Section 2, an independent check. Adding h^2 gives the middle row of the Hodge diamond

$$(h^{4,0}, h^{3,1}, h^{2,2}, h^{1,3}, h^{0,4}) = (0, 120, 581, 120, 0), \qquad b_4 = 821.$$

Remark 4.2 (What is and is not formalised)**.** Lean verifies the *counting*: the coefficient list of $(1+t+t^2+t^3)^6$ and its entries at the Griffiths degrees. The identification of $\dim R_k$ with a Hodge number is Griffiths' theorem, which is cited. In particular $h^{3,1} = 120 \neq 0$; only $h^{4,0}$ (and $h^{0,4}$) vanish.

5 Computing the Hilbert series in Lean

Lean represents a polynomial in t by its coefficient list. The definitions (file `Hodge/FermatCubicHost.lean`) are

```
def conv (p q : List Nat) : List Nat :=
  (List.range (p.length + q.length - 1)).map fun k =>
    ((List.range (k + 1)).map fun i => p.getD i 0 * q.getD (k - i) 0).sum
def jacobianSeries (d : Nat) : List Nat :=
  (List.range 6).foldl (fun acc _ => conv acc (List.replicate (d - 1) 1)) [1]
def primHodge (d q : Nat) : Nat := (jacobianSeries d).getD (d * (q + 1) - 6) 0
```

`conv` is the Cauchy product, so `jacobianSeries d` is the coefficient list of $(1+t+\cdots+t^{d-2})^6$, the Hilbert series of $\mathbb{C}[x_0,\dots,x_5]/(x_0^{d-1},\dots,x_5^{d-1})$, and `primHodge d q` is its coefficient in Griffiths degree $d(q+1)-6$. For $d=5$ the kernel evaluates these closed terms by `decide`: `quintic_series` (length 19, sum 4096) and `quintic_primitive_hodge` $(120, 580, 120)$. The same functions give $(1, 20, 1)$ for the cubic fourfold and $(426, 1751, 426)$ for the sextic fourfold (`cubic_primitive_hodge`, `sextic_primitive_hodge`), which are the known values and serve as sanity checks.

What the list computation does not prove. Lean does not formalise the isomorphism between the quotient ring $K[x]/J$ and the tensor product, nor its Hilbert series as a statement about a Mathlib graded algebra; the link is the elementary monomial-basis argument of Section 3. For $\psi = 2$ the identification uses that the six partials of a smooth quintic form a regular sequence (cited), and is confirmed by Singular.

6 The standalone certificate repository

`https://github.com/BenFrohman/FermatQuinticJacobian` contains `FermatQuinticJacobian.lean`, which defines $G \in \mathbb{Z}[x_0,\dots,x_5]$ as a Mathlib `MvPolynomial` and proves `Hodge.FermatQuintic.partial_off` ($\partial_j x_i^5 = 0$ for $j \neq i$), `partial_on` ($\partial_i x_i^5 = 5x_i^4$) and `partial_eq` ($\partial_i G = 5x_i^4$). Its CI audits these 3 declarations: each depends on `[propext, Classical.choice, Quot.sound]`.

7 Singular members have infinite Jacobian rings

For X_1 the partials have the common zero $p = (1,1,1,1,1,0)$ (Lean), so $K[x]/J(G_1)$ is not Artinian and there is no Griffiths description; this is why the Hodge numbers above are claimed only for smooth members. For X_2, Lean's `DworkQuinticPsi2.smooth` says the partials have only the trivial common zero, so $J(G_2)$ is generated by a regular sequence and $R(G_2)$ has the same Hilbert series as the Fermat case, which Singular confirms degree by degree.

8 What is proved, computed, and cited

Statement	Status
$\partial_i G = 5x_i^4$; partials vanish together only at 0 (char. 0)	Lean
Coefficient list of $(1+t+t^2+t^3)^6$: length 19, sum 4096, entries 120, 580, 120	Lean (`decide`)
$R(G)$ has that Hilbert series	elementary (monomial basis)
Same series for every smooth quintic	cited (regular sequence)
$h^{4-q,q}_{\mathrm{prim}} = \dim R_{5q-1}$	cited (Griffiths)
X_1 singular; X_2 smooth	Lean
Hilbert function of $R(G_2)$	Singular (exact over $\mathbb{Q}$)
X_2 contains no plane; ≥ 240 Hodge classes	Singular; bookkeeping + cited theory
Hodge conjecture for X_2	**open**

9 The Dwork-type pencil

Consider the pencil

$$G_\psi = \sum_{i=0}^{5} x_i^5 - 5\psi\, x_0x_1x_2x_3x_4, \qquad X_\psi = V(G_\psi) \subset \mathbb{P}^5.$$

The product omits x_5, so every member is homogeneous of degree 5 (an earlier proposal using $x_0 \cdots x_5$ is *not* homogeneous and does not define a hypersurface in $\mathbb{P}^5$; this is also recorded in Lean as `FermatQuintic.G_not_euler_degree5`). X_0 is the Fermat quintic.

9.1 $\psi = 1$ is singular

Proposition 9.1 (Lean: `DeformedQuinticHost.singular_at_p`, `not_smooth`)**.** *The point $p = [1:1:1:1:1:0]$ lies on X_1 and all partial derivatives of G_1 vanish at p. Hence X_1 is singular.*

Proof. $G_1(p) = 5 - 5 = 0$; $\partial_i G_1 = 5x_i^4 - 5\prod_{j\leq 4, j\neq i} x_j$ for $i \leq 4$ equals $5 - 5 = 0$ at p; and $\partial_5 G_1 = 5x_5^4 = 0$. □

More generally X_ψ is singular when $\psi^5 = 1$ (the points $[\zeta^{a_0} : \cdots : \zeta^{a_4} : 0]$); only $\psi = 1$ is formalised. The Griffiths machinery of Section 4 does not apply to X_1.

9.2 $\psi = 2$ is smooth

Theorem 9.2 (Lean: `DworkQuinticPsi2.smooth`)**.** *Over a field K of characteristic 0, the partials of $G_2 = \sum x_i^5 - 10\, x_0 \cdots x_4$ vanish simultaneously only at 0. Hence X_2 is a smooth quintic fourfold.*

Proof. The partials are $5x_5^4$ and $5x_i^4 - 10\prod_{j\leq 4, j\neq i} x_j$ ($i \leq 4$). So $x_5 = 0$ and $x_i^4 = 2\prod_{j\neq i} x_j$. Put $P = x_0x_1x_2x_3x_4$. Multiplying the five relations, $P^4 = \prod x_i^4 = 2^5 \prod_i \prod_{j\neq i} x_j = 32P^4$, so $31P^4 = 0$ and $P = 0$. Then $x_i^5 = x_i \cdot 2\prod_{j\neq i} x_j = 2P = 0$ for each $i \leq 4$. This is exactly the Lean proof (it uses `linear_combination` for each step). □

By Theorem 9.2, Section 4 applies verbatim to X_2: primitive Hodge numbers $0, 120, 580, 120, 0$ (Lean arithmetic: `dwork_primitive_hodge`). Independently, Singular computed the Hilbert function of $\mathbb{Q}[x]/J(G_2)$ as $1, 6, 21, 56, 120, 216, 336, 456, 546, 580, 546, \ldots, 1$ (total 4096), matching Table 1 (script `scripts/dwork_quintic_psi2/hilb.sing`).

9.3 Computations outside Lean for $\psi = 2$

- **No planes.** A plane in $\mathbb{P}^5$ is the row space of a 3×6 matrix in reduced form. S_5 (permuting $x_0, \ldots, x_4$) preserves G_2, so the 20 Grassmannian charts reduce to two orbit representatives (pivots $\{0,1,2\}$ and $\{0,1,5\}$). Restricting G_2 gives 21 equations in 9 unknowns; a Gröbner basis over $\mathbb{Q}$ is $\{1\}$, so X_2 contains no plane over $\mathbb{C}$. The same code finds the planes of the Fermat quintic (sanity check). (Script `planes.sing`; exact, not Lean.)

- **Hodge classes.** Let $H = \{a \in (\mathbb{Z}/5)^6 : a_0 + \cdots + a_4 \equiv 0\}$/diagonal (order 625) act by $x_i \mapsto \zeta^{a_i} x_i$; it preserves every G_ψ. Deforming to $\psi = 0$, the 820 Fermat characters group into 500 H-eigenspaces (240 of rank 1, 200 of rank 2, 60 of rank 3). The 240 rank-1 eigenspaces are, with all Galois conjugates, of type $(2,2)$, so they give *at least 240 primitive rational Hodge classes on X_ψ for every smooth ψ*, including $\psi = 2$ (script `chars.py`; bookkeeping plus cited theory). The exact rank at $\psi = 2$ is not computed.

9.4 Status of the Hodge conjecture for X_2: open

For the Fermat quintic the Hodge conjecture is known (Shioda [2], Ran [3]: prime degree), with cycles built from linear subspaces. At $\psi = 2$ those planes do not deform (X_2 contains none), and G_2 has 7 monomials in 6 variables, so it is not of Delsarte type. Showing that the 240 classes above stay algebraic along the pencil is an instance of the *variational Hodge conjecture*, which is open. We found no published result for $\psi = 2$. **No claim of the Hodge conjecture for X_2 is made.**

10 Lean formalisation

10.1 Where the proofs live

All machine-checked statements cited in this paper are in the repository `https://github.com/BenFrohman/HODGE` (Lean 4 `v4.22.0`, Mathlib `v4.22.0`), in three files:

- `Hodge/QuinticHost.lean` (Fermat quintic, merged in PR #26);
- `Hodge/DeformedQuinticHost.lean` (Dwork member $\psi = 1$, PR #26);
- `Hodge/DworkQuinticPsi2.lean` (Dwork member $\psi = 2$, PR #28).

The counting functions `conv`, `jacobianSeries` and `primHodge` are defined in `Hodge/FermatCubicHost.lean` and reused. The polynomial is an honest `MvPolynomial (Fin 6) K` over an arbitrary field `K`; partial derivatives are Mathlib's `MvPolynomial.pderiv`. None of the files uses `sorry`, `native_decide` or a user-declared `axiom`; the repository CI enforces this with a source scan.

10.2 Theorem inventory

Lean declaration	Statement
`QuinticHost.G_isHomogeneous`	$G = \sum_{i=0}^{5} x_i^5$ is homogeneous of degree 5.
`QuinticHost.pderiv_G`	$\partial G/\partial x_i = 5x_i^4$.
`QuinticHost.smooth`	If $\mathrm{char}\, K = 0$ and all $\partial_i G(x) = 0$ then $x = 0$.
`QuinticHost.quintic_series`	The coefficient list of $(1+t+t^2+t^3)^6$ has length 19 and sum 4096.

`QuinticHost.quintic_primitive_hodge`	$R_4 = 120$, $R_9 = 580$, $R_{14} = 120$.
`QuinticHost.quintic_h40_zero`	$5 \cdot 1 < 6$ (so the Griffiths degree for $h^{4,0}$ is negative).
`QuinticHost.quintic_h31_ne_zero`	$R_4 \neq 0$.
`DeformedQuinticHost.G_isHomogeneous`	$G_1 = \sum x_i^5 - 5x_0x_1x_2x_3x_4$ is homogeneous of degree 5.
`DeformedQuinticHost.singular_at_p`	$G_1(p) = 0$ and $\partial_i G_1(p) = 0$ for all i, $p = (1, 1, 1, 1, 1, 0)$.
`DeformedQuinticHost.not_smooth`	The Jacobian criterion fails for G_1.
`DworkQuinticPsi2.G_isHomogeneous`	$G_2 = \sum x_i^5 - 10x_0x_1x_2x_3x_4$ is homogeneous of degree 5.
`DworkQuinticPsi2.eval_pderiv`	Explicit formulas for the six partials of G_2.
`DworkQuinticPsi2.smooth`	If $\mathrm{char}\, K = 0$ and all $\partial_i G_2(x) = 0$ then $x = 0$.
`DworkQuinticPsi2.psi_not_root`	$2^5 \neq 1$.
`DworkQuinticPsi2.dwork_primitive_hodge`	$R_4 = 120$, $R_9 = 580$, $R_{14} = 120$ for the series $(1 + t + t^2 + t^3)^6$.

10.3 Axiom audit

The HODGE CI runs `lake env lean AxiomAudit.lean` (one `#print axioms` per declaration) and fails unless every declaration depends only on Lean's standard axioms `propext`, `Classical.choice`, `Quot.sound`, or on none. Output on `main` (the $\psi = 2$ declarations were added to the audit in HODGE PR #31):

Declarations	`#print axioms`
`QuinticHost.{G_isHomogeneous, pderiv_G, smooth}`	`[propext, Classical.choice, Quot.sound]`
`QuinticHost.{quintic_series, quintic_primitive_hodge, quintic_h31_ne_zero}`	`[propext]`
`QuinticHost.quintic_h40_zero`	none
`DeformedQuinticHost.*` (3 declarations)	`[propext, Classical.choice, Quot.sound]`
`DworkQuinticPsi2.{G_isHomogeneous, eval_pderiv, smooth}`	`[propext, Classical.choice, Quot.sound]`
`DworkQuinticPsi2.{psi_not_root, dwork_primitive_hodge}`	`[propext]`

`Classical.choice` enters through Mathlib's polynomial and field library, not through any step specific to this paper. The counting theorems are closed by `decide` and use only `propext`.

11 Reproducibility

- Build: `git clone https://github.com/BenFrohman/HODGE; cd HODGE; lake exe cache get; lake build`.
- Axiom audit: `lake env lean AxiomAudit.lean`; CI workflow `Lean certificate`.
- Singular scripts for $\psi = 2$: `scripts/dwork_quintic_psi2/` (`hilb.sing`, `planes.sing`, `fermat.sing`, `chars.py`) with outputs committed next to them.
- Standalone certificate repository: `https://github.com/BenFrohman/FermatQuinticJacobian`, which carries its own CI with the same axiom check.

Scope

This paper proves smoothness statements and Jacobian-ring counts in Lean and cites Griffiths' theorem for the Hodge-theoretic interpretation. It makes no claim about algebraic cycles on the quintic fourfold beyond the cited Fermat results, and it does not claim the Hodge conjecture for any non-Fermat member of the pencil.

References

[1] P. A. Griffiths, On the periods of certain rational integrals I, II, *Ann. of Math.* 90 (1969), 460–541.

[2] T. Shioda, The Hodge conjecture for Fermat varieties, *Math. Ann.* 245 (1979), 175–184.

[3] Z. Ran, Cycles on Fermat hypersurfaces, *Compositio Math.* 42 (1980), 121–142.

[4] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I, II*, Cambridge Univ. Press, 2002–2003.

[5] The mathlib Community, The Lean mathematical library, *CPP 2020*, 367–381.